

MISCELLANEOUS PROTOCOLS

39.1 EFI Timestamp Protocol

39.1.1 EFI_TIMESTAMP_PROTOCOL

Summary

The Timestamp protocol provides a platform independent interface for retrieving a high resolution timestamp counter.

GUID

```
#define EFI_TIMESTAMP_PROTOCOL_GUID \
{ 0xafbfde41, 0x2e6e, 0x4262, \
  { 0xba, 0x65, 0x62, 0xb9, 0x23, 0x6e, 0x54, 0x95 } }
```

Protocol Interface Structure

```
typedef struct _EFI_TIMESTAMP_PROTOCOL {
    TIMESTAMP_GET           GetTimestamp;
    TIMESTAMP_GET_PROPERTIES GetProperties;
} EFI_TIMESTAMP_PROTOCOL;
```

39.1.2 EFI_TIMESTAMP_PROTOCOL.GetTimestamp()

Summary

Retrieves the current timestamp counter value.

Prototype

```
typedef
UINT64
(EFIAPI *TIMESTAMP_GET) (
    VOID
);
```

Description

Retrieves the current value of a 64-bit free running timestamp counter.

The counter shall count up in proportion to the amount of time that has passed. The counter value will always roll over to zero. The properties of the counter can be retrieved from GetProperties().

The caller should be prepared for the function to return the same value twice across successive calls. The counter value will not go backwards other than when wrapping, as defined by EndValue in GetProperties().

The frequency of the returned timestamp counter value must remain constant. Power management operations that affect clocking must not change the returned counter frequency. The quantization of counter value updates may vary as long as the value reflecting time passed remains consistent.

Return Value

The current value of the free running timestamp counter.

39.1.3 EFI_TIMESTAMP_PROTOCOL.GetProperties()

Summary

Obtains timestamp counter properties including frequency and value limits.

Prototype

```
typedef
EFI_STATUS
(EFIAPI *TIMESTAMP_GET_PROPERTIES) (
    OUT    EFI_TIMESTAMP_PROPERTIES    *Properties
);
```

Parameters

Properties

The properties of the timestamp counter. See “Related Definitions” below.

Description

Retrieves the timestamp counter properties structure.

Related Definitions

```
typedef struct {
    UINT64      Frequency;

    UINT64      EndValue;
} EFI_TIMESTAMP_PROPERTIES;
```

Frequency

The frequency of the timestamp counter in Hz.

EndValue

The value that the timestamp counter ends with immediately before it rolls over. For example, a 64-bit free running counter would have an EndValue of 0xFFFFFFFFFFFFFFFF. A 24-bit free running counter would have an EndValue of 0xFFFFFF.

Status Codes Returned

EFI_SUCCESS	The properties were successfully retrieved.
EFI_DEVICE_ERROR	An error occurred trying to retrieve the properties of the timestamp counter subsystem. <i>Properties</i> is not updated.
EFI_INVALID_PARAMETER	<i>Properties</i> is NULL .

39.2 Reset Notification Protocol

39.2.1 EFI_RESET_NOTIFICATION_PROTOCOL

Summary

This protocol provides services to register for a notification when `ResetSystem` is called.

GUID

```
#define EFI_RESET_NOTIFICATION_PROTOCOL_GUID \
{ 0x9da34ae0, 0xeaf9, 0x4bbf, \
  { 0x8e, 0xc3, 0xfd, 0x60, 0x22, 0x6c, 0x44, 0xbe } }
```

Protocol Interface Structure

```
typedef struct _EFI_RESET_NOTIFICATION_PROTOCOL {
    EFI_REGISTER_RESET_NOTIFY      RegisterResetNotify;
    EFI_UNREGISTER_RESET_NOTIFY    UnRegisterResetNotify;
} EFI_RESET_NOTIFICATION_PROTOCOL;
```

Parameters

RegisterResetNotify

Register a notification function to be called when `ResetSystem()` is called.

UnRegisterResetNotify

Removes a reset notification function that has been previously registered with `RegisterResetNotify()`.

39.2.2 EFI_RESET_NOTIFICATION_PROTOCOL.RegisterResetNotify()

Summary

Register a notification function to be called when `ResetSystem()` is called.

Prototype

```
typedef
EFI_STATUS
(EFIAPI *EFI_REGISTER_RESET_NOTIFY) (
    IN EFI_RESET_NOTIFICATION_PROTOCOL    *This,
    IN EFI_RESET_SYSTEM                   *ResetFunction,
);
```

Parameters

This

A pointer to the *EFI_RESET_NOTIFICATION_PROTOCOL* instance.

ResetFunction

Points to the function to be called when a *ResetSystem()* is executed.

Description

The *RegisterResetNotify()* function registers a notification function that is called when *ResetSystem()* is called and prior to completing the reset of the platform.

The registered functions must not perform a platform reset themselves. These notifications are intended only for the notification of components which may need some special-purpose maintenance prior to the platform resetting.

The list of registered reset notification functions are processed if *ResetSystem()* is called before *ExitBootServices()* . The list of registered reset notification functions is ignored if *ResetSystem()* is called after *ExitBootServices()* .

Status Codes Returned

EFI_SUCCESS	The reset notification function was successfully registered.
EFI_INVALID_PARAMETER	ResetFunction is NULL .
EFI_OUT_OF_RESOURCES	There are not enough resources available to register the reset notification function.
EFI_ALREADY_STARTED	The reset notification function specified by ResetFunction has already been registered.

39.2.3 EFI_RESET_NOTIFICATION_PROTOCOL.UnregisterResetNotify()

Summary

Unregister a notification function.

Prototype

```
typedef
EFI_STATUS
(EFIAPI *EFI_UNREGISTER_RESET_NOTIFY) (
    IN EFI_RESET_NOTIFICATION_PROTOCOL    *This,
    IN EFI_RESET_SYSTEM                  *ResetFunction
);
```

Parameters

This

A pointer to the EFI_RESET_NOTIFICATION_PROTOCOL instance.

ResetFunction

The pointer to the ResetFunction being unregistered.

Description

The UnregisterResetNotify() function removes the previously registered notification using RegisterResetNotify().

Status Codes Returned

EFI_SUCCESS	The reset notification function was unregistered.
EFI_INVALID_PARAMETER	ResetFunction is NULL .
EFI_INVALID_PARAMETER	The reset notification function specified by ResetFunction was not previously registered using RegisterResetNotify().